

CIND820: Big Data Analytics Project

Final Report

Dealership Vehicle Dataset

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GitHub Repository

https://github.com/Logic9397/Vehicle_Dataset_for_CIND820/tree/main

Introduction:

The automotive industry continues to undergo significant transformation, emphasized by growing consumer purchasing and the increasing competition between fuel alternatives. As the variety of vehicles expand, so too does the complexity of accurately determining vehicle pricing. This study employs a machine learning framework to analyze, classify, and model vehicle pricing behavior, thereby contributing to the fields of predictive analytics and pattern recognition within the automotive market.

The aim of this research is to examine the relationship between various vehicle attributes and their market valuations, utilizing a dataset consisting of new vehicles. Specifically, the study seeks to address the following research questions:

- **Classification and Clustering:** Is it possible to categorize vehicles into distinct segments such as budget, mid-range, and luxury?
- **Feature Contribution Analysis:** How does the Manufacturer's Suggested Retail Price (MSRP) correlate with features such as engine type, make, drivetrain, fuel type, and trim level? Which attributes have the most significant influence on vehicle pricing?
- **Price Distribution Analysis:** What patterns emerge when analyzing the distribution of new vehicle prices in increments of \$5,000 or \$10,000?
- **Consumer Preference Trends:** What are the most frequently selected combinations of exterior and interior colors among new vehicle listings?
- **Predictive Modeling:** Can an accurate predictive model be developed using machine learning techniques to estimate vehicle prices based on available attributes?

These research objectives are designed to identify meaningful patterns in automotive pricing, improve the understanding of market segmentation, and establish a data-driven framework for forecasting vehicle prices.

Literature Review:

Lapatta, N. T., & Husin, A. (2024). *Predicting potential car buyers using logistic regression algorithm.* *Sistemasi: Jurnal Sistem Informasi*, 13(3), 1147–1156. <http://sistemasi.ftik.unisi.ac.id>

Lapatta et al provide a fantastic source by analyzing a similar vehicle dataset using logistic regression. Previous attempts have used other modelling methods such as Naive Bayes and ARIMA, but they have focused on sales. Instead Lapatta et al appear to use logistic regression as a means to predict consumer behaviour. Their final logistic model incorporating factors such as income, age, gender and marital status reached 95% accuracy.

Kavitha, V., Sai, P. D., Revathi, S., & Reddy, P. V. K. (2019). Car buying decision using machine learning algorithms. *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 7(IV), 293–298. <https://doi.org/10.22214/ijraset.2019.4494>

Kavitha et al provide a good high level overview of various machine learning models used to predict car purchasing. The group tried using logistic regression, K-Nearest Neighbor and decision tree classification. Of the three, logistic regression performed poorly and decision tree classifier performed the best with a 96.5% accurate model.

Ledesma-Alonso, R., & Becerra-Nuñez, G. (2024). Electric or gasoline: A simple model to decide when buying a new vehicle. *Environmental Research Communications*, 6(2), 025015. <https://doi.org/10.1088/2515-7620/ad2949>

Ledesma-Alonso et al build a non-machine learning model using simple linear mathematics to predict breakeven points of car ownership and purchase price. They focus on four consumer driven behaviours including: driving habits, economic trends, duration of car ownership and vehicle depreciation. They conclude that electric vehicles are better for high mileage trips in congested areas, while gasoline vehicles are better for low mileage and light traffic areas.

Ong, A. K. S., Cordova, L. N. Z., Longanilla, F. A. B., Caprecho, N. L., Javier, R. A. V., Borres, R. D., & German, J. D. (2023). Purchasing intentions analysis of hybrid cars using random forest classifier and deep learning. *World Electric Vehicle Journal*, 14(8), 227. <https://doi.org/10.3390/wevj14080227>

Ong et al provide a resource for modelling Filipino car purchasing behaviour towards hybrid vehicles. They use a variety of machine learning models to gauge accuracy including: decision tree, random forest and deep learning neural network. Their findings concluded that deep learning neural networks provided the most accurate model at 96.60%, with random forest classifiers coming second at 94% accuracy. For CIND820, deep learning neural networks may be beyond the scope of the course, but random forest classifiers could be investigated.

Amik, F. R., Lanard, A., Ismat, A., & Momen, S. (2021). *Application of machine learning techniques to predict the price of pre-owned cars in Bangladesh.* Information, 12(12), 514. <https://doi.org/10.3390/info12120514>

Amik et al built a machine learning model to predict the price of pre-owned vehicles in Bangladesh. They covered a variety of machine learning models including: linear regression, LASSO regression, decision tree, random forest and XGBoost. Among the five machine learning methods tested, XGBoost generated the highest accuracy with a R-squared value of 91.32%. The second best model was a random forest model with an R-squared value of 90.14%. For CIND820, random forest appears to be performing well among peer-reviewed sources.

Al-Turjman, F., Hussain, A. A., Alturjman, S., & Altrjman, C. (2022). *Vehicle price classification and prediction using machine learning in the IoT smart manufacturing era.* Sustainability, 14(15), 9147. <https://doi.org/10.3390/su14159147>

Al-Turjman et al focus on testing machine learning models to predict vehicle prices based on the features of those vehicles. They test four machine learning models including: linear regression, support vector machine (SVM), decision tree and neural networks. Unfortunately, the models alone performed quite poorly and an “ensemble” or combination of machine learning models was required. Standalone, the models performed poorly with SVM performing the best at 49% accuracy. The group used a combination of models that ended up with a SVM ensemble model predicting with 90% accuracy.

Data Description:

I have selected a vehicle dataset consisting of new vehicles available for purchase. The dataset consists of 18 variables for each vehicle in the table below with a total 1002 data points.

Variable	Description	Data Type
name	Name of vehicle	string
description	Description given by dealership and highlighted features	string
make	Make / brand of vehicle	string
model	Model of vehicle	string
type	New vehicle or used vehicle (Dataset only contains new)	string
year	Model year of vehicle	integer
price	Dealership price of new vehicle	float
engine	Model of engine in vehicle	string
cylinders	Number of cylinders (if applicable)	float
fuel	Fuel source of vehicles (Gasoline, electric etc.)	string
mileage	Number of miles on the vehicle odometer	float
transmission	Vehicle transmission	string
trim	Vehicle trim level for model	string
body	Vehicle body type	string
doors	Number of doors on the vehicle	float
exterior	Exterior paint color of the vehicle	string
interior	Interior paint color of the vehicle	string
drivetrain	Drivetrain of the vehicle (All wheel drive, front wheel drive etc.)	string

Approach:



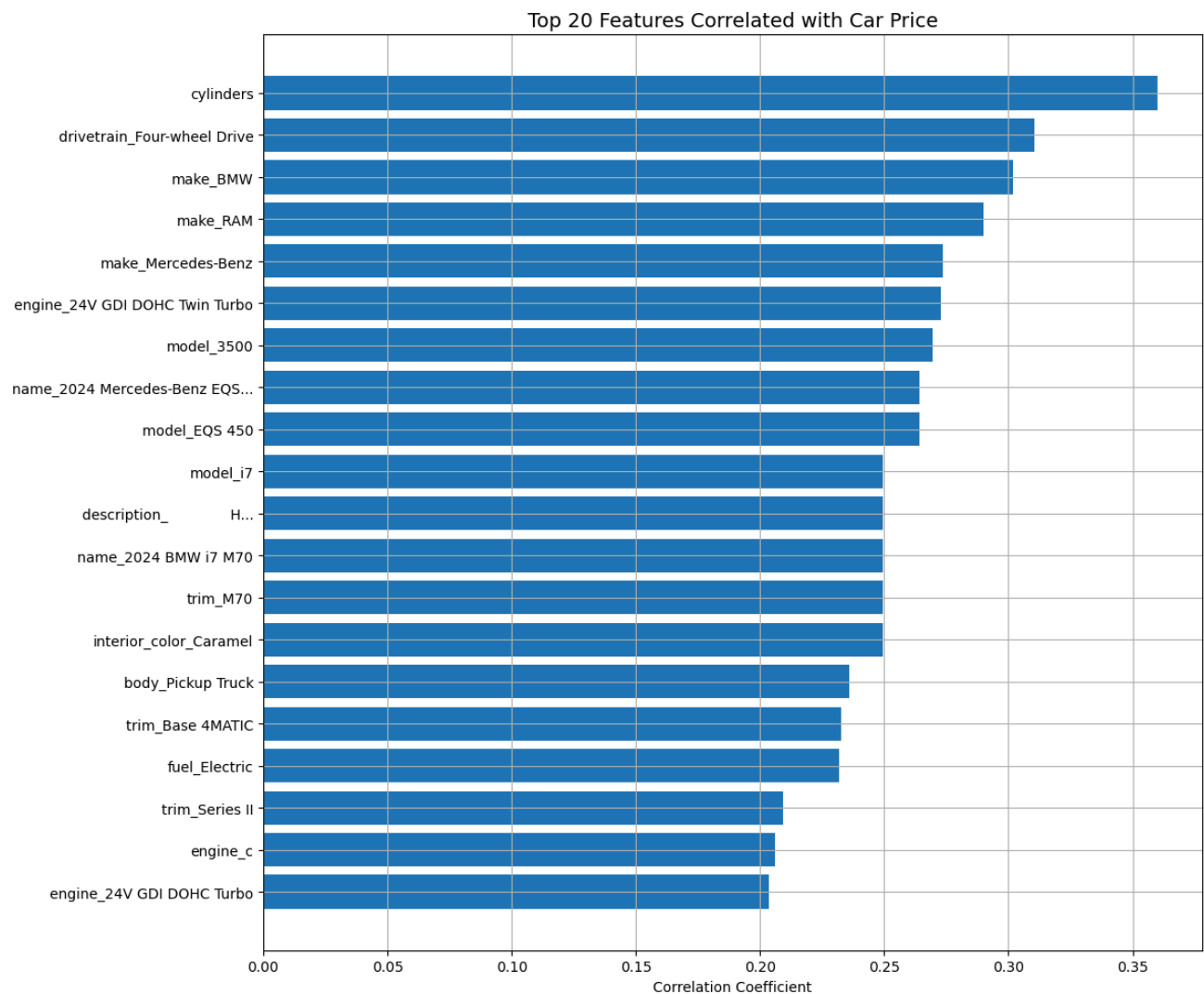
The process starts with pre-processing the vehicle data to handle abnormal and missing data. There were no outliers present in this data set, but there were several missing values scattered among the data. My approach to resolving these missing values are below.

name (0)	N/A
description (56)	Substitute null values with a placeholder such as "No Description"
make (0)	N/A
model (0)	N/A
type (0)	N/A
year (0)	N/A
price (23)	Drop rows missing price, as price is the dependent variable for modelling
engine (2)	Substitute null values with a placeholder such as "Unknown Engine"
cylinders (105)	Substitute median value to limit outliers
fuel (7)	Substitute mode value as the most common fuel type
mileage (34)	Since mileage varies in this dataset, median limits skewing the data
transmission (2)	Substitute mode value as the most common transmission type
trim (1)	Substitute null values with a placeholder such as "Unknown Trim"
body (3)	Substitute mode value as the most common body type
doors (7)	Substitute mode value as the most common door configuration
exterior (5)	Substitute null values with a placeholder such as "Unknown"
interior (38)	Substitute null values with a placeholder such as "Unknown"
drivetrain (0)	N/A

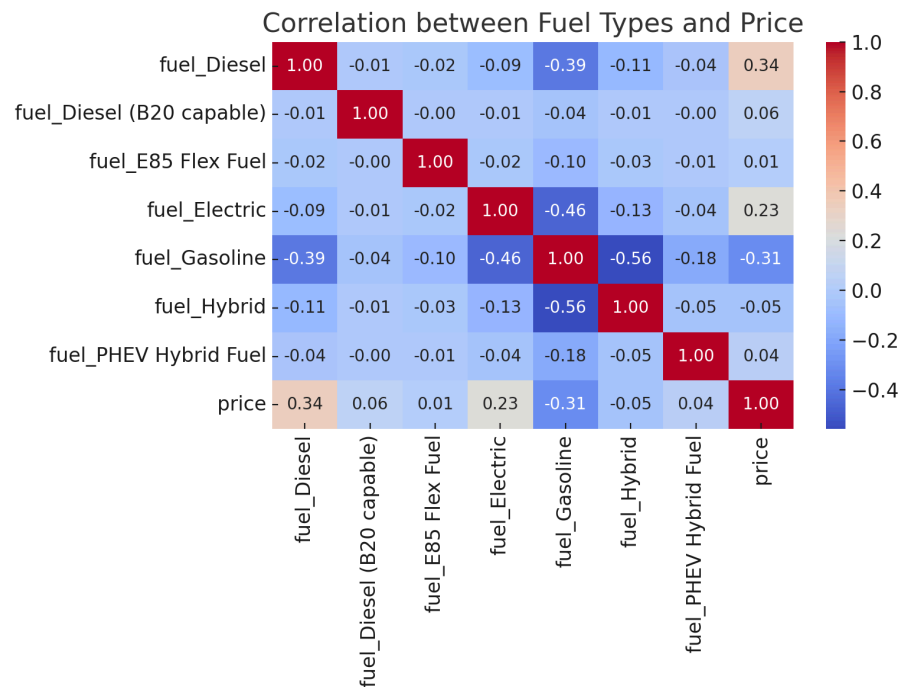
Next we perform a series of exploratory data analysis using univariate and bi-variate analysis to understand any initial patterns.

The below chart plots the top 20 features that correlate to vehicle price with the correlation coefficient plotted on the horizontal axis. Initial takeaways include:

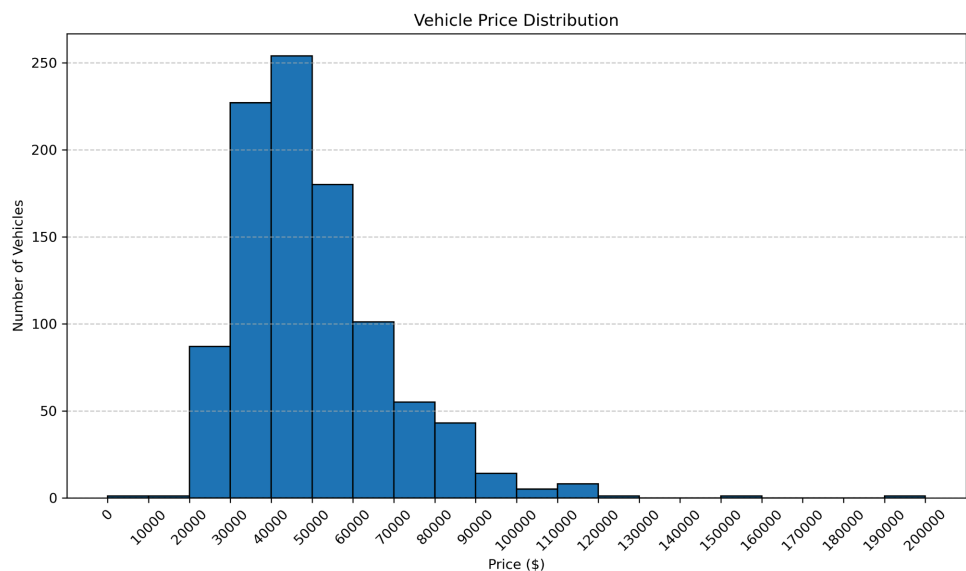
- The number of cylinders in a car engine correlates the most to the price.
- Mentions of “four wheel drive”, “model_RAM”, “model_3500” and “body_Pickup Truck” would imply that large sport utility vehicles or trucks correlate with higher prices.
- Luxury mentions such as “make_BMW”, “make_Mercedes-Benz” and “model_i7” also imply that luxury vehicles are driving prices higher.



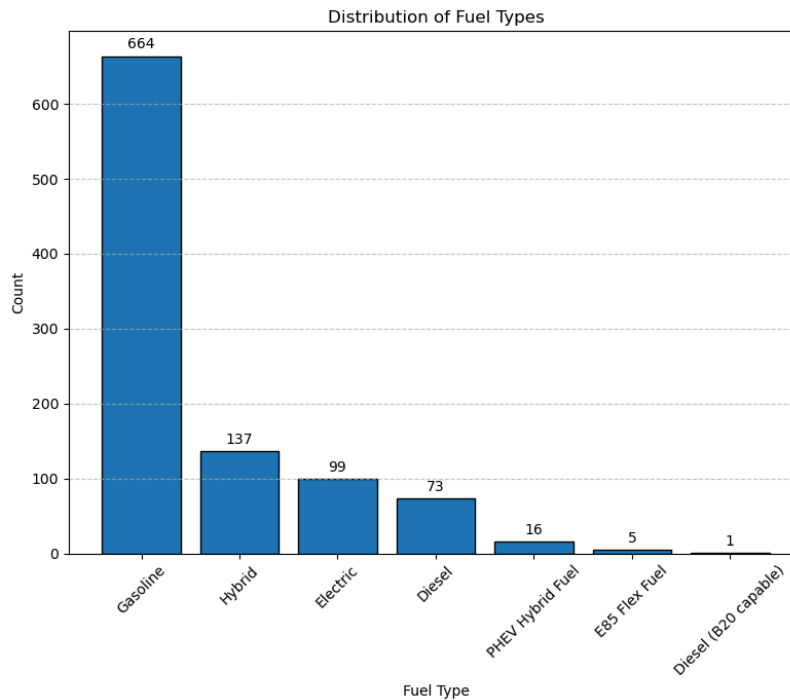
The below correlation heat map compares the various fuel types and overall price of the vehicle. Strangely, it appears that diesel vehicles correlate to higher prices with a positive correlation coefficient, whereas gasoline actually has a negative correlation coefficient implying lower prices.



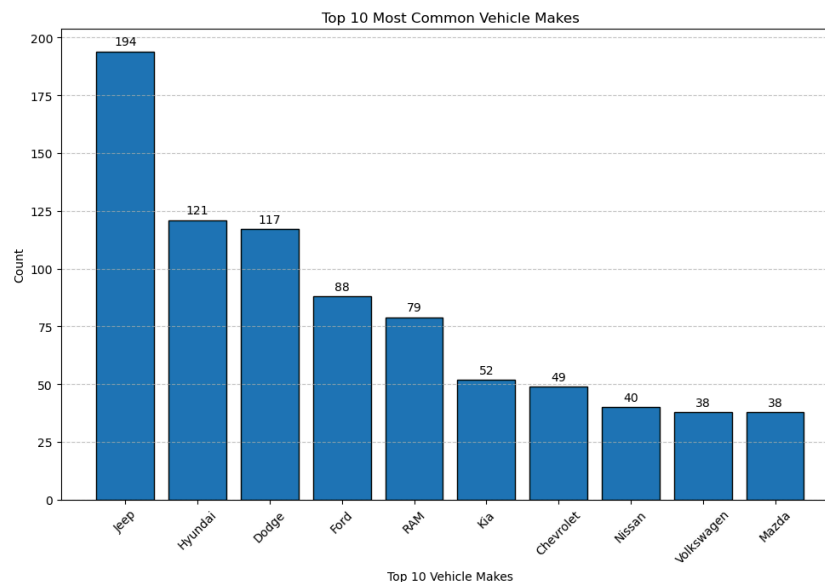
The below histogram categorizes the price of the vehicles into \$10,000 buckets. This gives a good visualization of the spread of the vehicle prices, and gives an idea of how the machine learning classification model may categorize vehicles in the future.



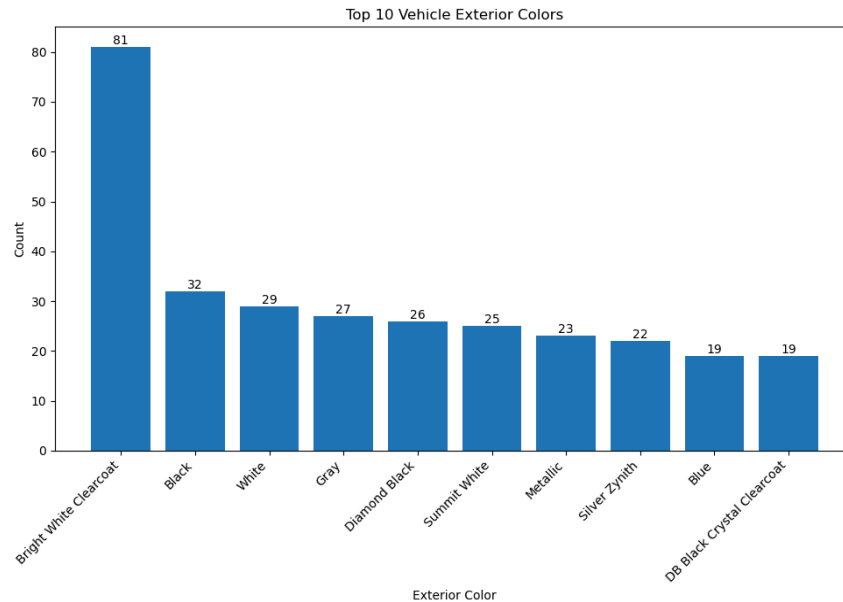
The below plot shows the distribution of fuel type within the data set. Since this raw data set was compiled between 2024 - 2025, we can make some inferences such as consumers moving towards hybrids over conventional gasoline vehicles. Hence the large spike in available gasoline vehicles. Interestingly, there is a lack of Plug-In Hybrid Electric Vehicles (PHEV) which I have a personal interest in, as they bridge a gap for consumers between gasoline and electric.



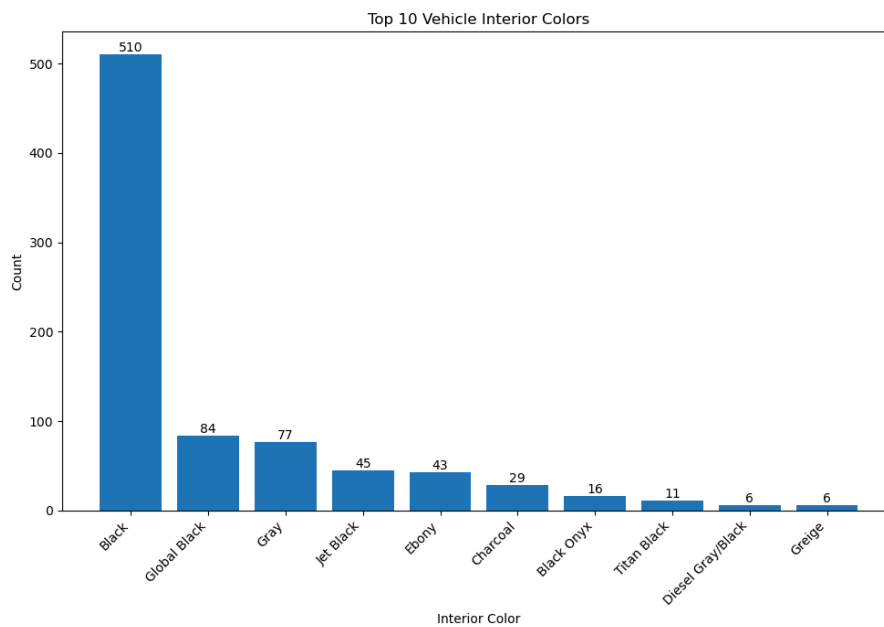
The below chart visualizes the raw count of vehicles by manufacturer make, listed from most frequent to least frequent. Since the original data set was a list of available vehicles from several dealerships, we might theorize that Jeep vehicles are not selling well. While popular makes such as Toyota or Honda are notably absent from the list.



The below chart describes the raw count of exterior color options within the data set and plots the colors from most to least frequent. It is not surprising to find that monochromatic colors such as black and grey constitute the majority of exterior colors. Monochromatic colors are easily manufacturable requiring no additional color additives and generally pleasing to most consumers.



The below chart describes the raw count of interior color options within the data set and plots the colors from most to least frequent. It is not surprising to find that monochromatic colors such as black and grey constitute the majority of interior colors. Monochromatic colors are easily manufacturable and contrast well with all exterior color combinations.



After initial exploratory analysis, we move onto the machine learning portion. Since I need to create models for both purchase categories (budget, luxury etc.) and price, I will need two models - one for classification and another for regression.

Based on peer-reviewed articles, the chosen algorithms that provided the most accurate models included:

- **Linear Regression** (Al-Turjman et al.)
- **Random Forest** (Amik et al.)
- **XGBoost** (Amik et al.)

For both regression and classification models, I made similar decisions to pre-process the data for evaluation. These include:

- Isolating the price as the model target
- Dropping redundant or irrelevant columns from the features:
 - price - price is the target parameter
 - name - redundant as make and model cover the name
 - description - not consistent across all vehicle models
 - type - all vehicles in the data set are new
 - engine - redundant as cylinders covers engine and engine description is not consistent across all vehicles
 - trim - trim models are not consistent across all vehicle models
- Train-test split is computed using 80-20 ratios.
- Cross-validation compared to train-test for classification models.

Starting with the classification models, accuracy was modelled with confusion matrices and computed as below:

Model: Linear Regression

Train-Test Accuracy: **0.73**

Cross-Validation Mean Accuracy: **0.69 (± 0.03)**

Model: Random Forest

Train-Test Accuracy: **0.77**

Cross-Validation Mean Accuracy: **0.73 (± 0.04)**

Model: XGBoost

Train-Test Accuracy: **0.71**

Cross-Validation Mean Accuracy: **0.71 (± 0.03)**

Similarly with the regression models, the model was computed as below :

Model: Random Forest

R² Score: **0.85**

Conclusion:

In conclusion, my recommendation for a machine learning model to accurately model the price in this vehicle data set is random forest for classification and regression.

The random forest algorithm produced a model with **77.0% train-test accuracy** and **73.0 +/- 4% accuracy with cross-validation**. This was the highest accuracy among the three algorithms when compared to both linear regression and XGBoost. Using the random forest model, a dealership would be able to market and price their vehicles according to the following classifications:

Class 0 (Budget): \$0.0 to \$36,600.0

Class 1 (Standard): \$36,600.0 to \$47,165.0

Class 2 (Premium): \$47,165.0 to \$58,909.0

Class 3 (Luxury): \$58,909.0 to \$195,895.0

Amik et al. were able to design a machine learning model around XGBoost, but had similarly successful performance with random forest, which my data set and model support.

Similarly when the random forest algorithm is applied as a regression model, the machine learning model is calculated with a R^2 score of 0.85. This presents a model that is fairly capable of predicting sale price for the dealership.

Both the classification and regression models can be used to support this dealership business, with the classification model supporting from a marketing standpoint and the regression model helping to support finances.

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